

Homework 1 Group I

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Question 1

Let $s_i = 1 - 2y_i$. For $\lambda > 0$, the objective function is

$$f_\lambda(\theta) = \underbrace{\sum_{i=1}^m \phi(s_i \langle \tilde{x}_i, \theta \rangle)}_{f(\theta)} + \frac{\lambda}{2} \|\theta\|^2, \quad \theta \in E.$$

We first show that f is convex. The smoothed hinge loss ϕ satisfies

$$\phi'(z) = \begin{cases} 0, & z \leq -1, \\ 1 + z, & -1 < z < 0, \\ 1, & z \geq 0. \end{cases}$$

The values and piecewise derivatives agree at $z = -1$ and $z = 0$. Thus ϕ is continuously differentiable. Its derivative is nondecreasing, so for any $a, b \in \mathbb{R}$,

$$\phi(b) - \phi(a) - \phi'(a)(b - a) = \int_a^b (\phi'(u) - \phi'(a)) du \geq 0.$$

(The inequality holds for either ordering of a and b .) By Theorem 4.21 of the lecture notes, ϕ is convex.

For any $\theta, \eta \in E$ and $t \in [0, 1]$, linearity of the inner product and convexity of ϕ give

$$\begin{aligned} \phi(s_i \langle \tilde{x}_i, (1-t)\theta + t\eta \rangle) &= \phi((1-t)s_i \langle \tilde{x}_i, \theta \rangle + ts_i \langle \tilde{x}_i, \eta \rangle) \\ &\leq (1-t)\phi(s_i \langle \tilde{x}_i, \theta \rangle) + t\phi(s_i \langle \tilde{x}_i, \eta \rangle). \end{aligned}$$

Summing over i yields $f((1-t)\theta + t\eta) \leq (1-t)f(\theta) + tf(\eta)$; hence f is convex.

Now

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex. Since $\lambda > 0$, Definition 4.5 of the lecture notes therefore implies that f_λ is λ -strongly convex. It is real valued on the whole Euclidean space E , so Theorem 4.28 applies: f_λ has a unique global minimizer θ^* .

Moreover, f_λ is convex and differentiable on E . Each loss term is a composition of the continuously differentiable function ϕ with a linear map, and the quadratic term is differentiable. By Corollary 4.23 of the lecture notes, a point is a global minimizer if and only if its gradient vanishes. So, what we can say about the critical points is that

$$\{\theta \in E : \nabla f_\lambda(\theta) = 0\} = \{\theta^*\}.$$

Question 2

We now set $\lambda = 0.005$. Write $g_i(\theta) = s_i \langle \tilde{x}_i, \theta \rangle$. Each $g_i : E \rightarrow \mathbb{R}$ is linear, hence differentiable, and ϕ is continuously differentiable as shown in Question 1. Thus $f = \sum_{i=1}^m \phi \circ g_i$ is differentiable. By additivity of the differential and the chain rule, for any $h \in E$,

$$Df(\theta)[h] = \sum_{i=1}^m \phi'(g_i(\theta)) Dg_i(\theta)[h].$$

Using the limit expression for the differential of g_i , we obtain

$$Dg_i(\theta)[h] = \lim_{t \rightarrow 0} \frac{g_i(\theta + th) - g_i(\theta)}{t} = \lim_{t \rightarrow 0} \frac{ts_i \langle \tilde{x}_i, h \rangle}{t} = s_i \langle \tilde{x}_i, h \rangle.$$

Substituting and using bilinearity of the inner product gives

$$\begin{aligned} Df(\theta)[h] &= \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \langle \tilde{x}_i, h \rangle \\ &= \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i, h \right\rangle. \end{aligned}$$

The quadratic term $r(\theta) = \frac{\lambda}{2} \|\theta\|^2$ is also differentiable. Indeed,

$$r(\theta + h) - r(\theta) - \lambda \langle \theta, h \rangle = \frac{\lambda}{2} \|h\|^2 = o(\|h\|),$$

so $Dr(\theta)[h] = \lambda \langle \theta, h \rangle$. Consequently, $f_\lambda = f + r$ is differentiable and

$$Df_\lambda(\theta)[h] = \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta, h \right\rangle.$$

By Definition 2.18 of the lecture notes, $Df_\lambda(\theta)[h] = \langle \nabla f_\lambda(\theta), h \rangle$. Since this equality holds for every $h \in E$, we conclude that

$$\nabla f_\lambda(\theta) = \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta.$$

Question 4

Given that ∇f_λ is L -Lipschitz continuous with constant $L = \sigma_{\max}(X)^2 + \lambda$ (where $\sigma_{\max}(X)$ is computed numerically), we select the constant step-size $\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda}$ as a consequence of Theorems 3.3 and 3.4. .

Question 7

The classification error is approximately 0.134% on the training set and 0.189% on the test set. The test error is slightly higher than the training error, as expected, but remains very small, indicating that the classifier generalizes well to unseen samples.